MIL-V-6753A
23 MARCH 1956
SUPERSEDING
MIL-V-6753
14 JUNE 1950

MILITARY SPECIFICATION

VOLTMETERS, 0-150 VOLT, 330 TO 1200 CYCLE, ALTERNATING CURRENT

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of 330 to 1200 cycle, 0–150V ac moving iron voltmeter.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-S-44 —Shock-Testing-Mechanism For Electrical-Indicating Instruments (2½ and 3½ Inch, Round, Flush Mounting, Panel-Type)

MIL-D-5028 —Drawings and Data Lists; Preparation of Manufacturers' (For Production Aircraft, Guided Missiles, Engines, Accessories and Other Auxiliary Equipment)

MIL-P-6064 —Packaging of Lightweight Aircraft Accessories

MIL-S-7742 —Screw Threads; Standard, Aeronautical

MIL-L-25142 —Luminescent Material, Fluorescent

STANDARDS

MILITARY

MIL-STD-129—Marking For Shipment and Storage

MIL-STD-130—Identification Marking of U. S. Military Property

MS33548 —Case, Instrument, 1⅞ Dial, Standard Dimensions For

MS33558 —Numerals and Letters —Aircraft Instrument Dial, Standard Form of

MS24332 —Voltmeter—0–150 Volt, 330 to 1200 Cycle, Alternating Current

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETINS

143 —Specifications and Standards; Use of

405 —Storage Life—Aeronautical Articles

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.—The voltmeter furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

3.2 Material

3.2.1 *Specifications and Standards.*—Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 142, except as provided in the following paragraph:

3.2.1.1 *Standard parts.*—AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.2 Protective treatment.—When materials are used in the construction of the voltmeter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.3 Fluorescent Luminescent Material.—Fluorescent luminescent material shall be in accordance with MIL–L–25142, type I or III, as applicable.

3.3 Design and Construction

3.3.1 The voltmeter shall be designed and constructed so that no parts will work loose in service.

3.3.2 The voltmeter shall be constructed so that adjustments and repairs can easily be made by personnel of operating units and overhaul bases without the use of special tools.

3.3.3 Operational range.—The range of operation for the voltmeter shall be from −54° to +71°C.

3.3.4 Case.—The case shall conform to Standard MS33548. The case shall be fabricated from metal.

3.3.5 Dial.—The dial shall be made of metal or of paper and shall be marked as shown on figure 1. Paper dials shall be satisfactory only when adequately bonded to and backed by a metal support, and when they meet all the tests specified herein without fading, discoloring, warping, or other deterioration.

3.3.6 Bearings and pivots.—Moving elements shall turn on hardened, accurately ground and highly polished steel pivots bearing on highly polished first quality sapphire, diamond jewels, or other materials that have been approved as an alternate to precious gems.

3.3.7 Pointers.—The body of the pointer shall be a maximum of $\frac{1}{16}$ inch wide. The entire visible portion of the pointer shall be marked with fluorescent luminescent material.

3.4 Interchangeability.—All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL–D–5028 shall govern changes in the manufacturer's part numbers.

3.5 Performance.—The voltmeter shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperatures.—Temperatures over the range of −54° to +71°C.

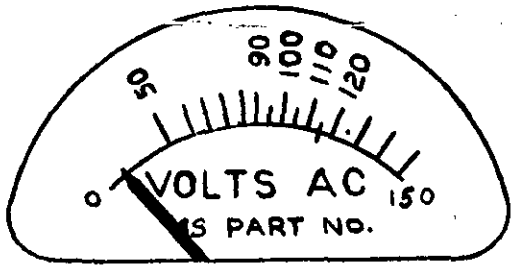
b. Humidity.—Relative humidity up to 100 percent.

c. Vibration.—Vibrations incident to service use. (See 3.5.11 and 4.5.8.)

d. Shock.—Shock forces as anticipated. (See 3.5.10 and 4.5.15.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

MIL-6753A



MARKING	HEIGHT OR LENGTH, INCH ±0.010	WIDTH OF LINE OR GRADUATION INCH ±0.005	FINISH
ALL NUMERALS LETTERING MAJOR GRADS SECTOR LINE BELOW SECTOR LINE AT 115 VOLTS GUADUATION	0.109 0.156 0.094 — 0.094	— — .030 .015 .030	FLUORESCENT LUMINESCENT MATERIAL
MINOR GRADS	0.062	.015	DURABLE DULL WHITE OR NATURAL METAL
MS PART NO.	0.062	—	DURABLE DULL BLACK

METER DIAL AND MARKING

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

3.5.1 Balancing.—When spring-controlled instruments are tilted 60° from the normal vertical position, in any direction, the pointer tip shall remain within 2 percent of total scale length at the zero point. Elements free to swing shall show a good mechanical balance. The balance shall be easily and readily adjustable.

3.5.2 Accuracy

3.5.2.1 Room temperature.—Unless otherwise specified, the meter shall indicate correctly within +2 percent of full scale deflection at room temperature of 25° to ±5°C.

3.5.2.2 Low temperature.—The change in reading when the meter is submitted to test at −55°C. shall not exceed 6 percent of full scale value.

3.5.2.3 Frequency.—The readings shall not change more than one percent of full scale value at frequencies of 300 to 1200 cycles, from the readings observed at 400 cycles.

3.5.3 Response time.—Response time shall not exceed 2.5 seconds when a voltage sufficient to give full scale momentary deflection is suddenly applied.

3.5.4 Compass effect.—The meters shall not cause a deflection of the compass of more than 6° when tested as specified in section 4.

3.5.5 Dielectric.—A voltage of 1000V rms, ac, at commercial frequency, applied for a period of 1 minute shall cause no breakdown of insulation.

3.5.6 Heat effect at 70°C.—Meters shall indicate freely after having been maintained at this temperature in heated air for 16 hours. The change in indication for each rise from 25°C to 70°C shall not exceed 5 times the temperature influence allowance during any of the three heat cycles when tested as specified in section 4. After the meters have been restored to 25°C to complete each cycle, the change in indication due to heat tests shall not exceed 2 percent for each cycle and 2 percent for the difference between the beginning of the first cycle (initial accuracy indication) and the end of the third cycle.

3.5.7 Low temperature.—The meter shall successfully withstand the test as specified in section 4.

3.5.8 Humidity resistance.—The meters shall operate satisfactorily after subjection to the humidity tests as specified in section 4. The permanent change in indication after the second humidity cycle shall not exceed 3 percent.

3.5.9 Sustained load capacity.—Meters shall withstand for 8 hours an applied load of 100 percent of rated capacity without a permanent change in indication of more than one percent of the scale length, without a permanent zero shift greater than 0.5 percent of the scale length, and without any permanent damage. The temporary zero shift shall not exceed 1.0 percent of the scale length.

3.5.10 Shock resistance.—Permanent errors in the indication resulting from shock of the character and intensity as specified in section 4, shall not exceed 2 percent of full scale deflection. No screws, bearing, pivots, or other parts shall become loosened or unduly damaged so as to affect the operation of the instrument.

3.5.11 Vibration.—Permanent errors in indication due to vibration of the character and intensity specified in section 4 shall not exceed 2 percent of full scale deflection. The friction error on the basis of the difference between ascending and descending readings shall not exceed 4 percent of full scale deflection. No screws, bearings, pivots or other parts shall be loosened or unduly damaged so as to affect the operation of the meter.

3.5.12 Mounting lugs.—Mounting lugs of the completely assembled case shall withstand for 1 minute without fracture, a load of 175 pounds applied along the mounting hole axis and toward the front of the meter.

3.6 Means of adjustment.—It shall be ascertained by suitable tests that the meters

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

have some practical shop means of adjustment to accuracy.

3.7 Pointer zero correction adjustment.—Zero correction adjustment shall be provided for not less than 3 percent of the scale length, above and below the zero position. The adjustment mechanism (or screw) shall be designed and so constructed to prevent damage when the meter is placed face down and also shall have sufficient friction to prevent it from shaking loose and changing its adjustment when subjected to the vibration tests specified herein. The range of zero adjustment shall be taken and recorded.

3.8 Bearing adjustment screw.—To prevent tampering, the bearing adjustment screw shall not be exposed.

3.9 Terminal posts.—The terminal posts shall be threaded and shall be provided with the proper nuts and lock washers for attaching cable terminals in accordance with MS24332. Plating of the posts shall be subject to the approval of the procuring activity.

3.10 Scale visibility.—The scale and transparent front shall be so designed that the full scale of the meter can be easily read at any angle up to 45° below or 30° on either side of the axis through the center of the meter perpendicular to the plane of mounting with illumination coming from above and in front and at an angle of 45° from the same axis. Under these conditions there shall be no objectionable shadows on the graduated scale.

3.11 Parallax.—To avoid parallax, the clearance between the pointer and the graduated scale shall not exceed 0.05 inch.

3.12 Damping.—The damping factor shall not be less than 1.5 for these meters.

3.13 Temperature influence.—The temperature influence when tested as specified in section 4 shall not exceed 0.75 percent for these meters.

3.14 Insulation resistance.—The insulation resistance between all live parts and the metal panel upon which the instrument is mounted shall be not less than 20 megohms at a potential of 500V dc.

3.15 Weight.—The weight of the meter shall not exceed 0.75 pound.

3.16 Scale (range and length).—The range of the scale shall be 0–150V. The scale length shall be not less than $1\frac{7}{16}$ inch as measured by the arc described by the pointer tip.

3.17 Adjustment.—The meter shall be adjusted to give full scale reading at 150V.

3.18 Sensitivity.—The meter shall have a sensitivity of 50 ohms per volt minimum.

3.19 Power consumption.—The power consumed by the meter shall be a maximum of 3 watts at full scale deflection.

3.20 Screw threads.—Unless otherwise specified, the threads of all machine screws 0.060 inch or larger shall be in accordance with MIL–S–7742.

3.21 Markings.—All markings shall be durable to withstand usage encountered in service.

3.21.1 Dial markings.—Dials shall be marked as shown on figure 1. The form of letters and numerals shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies. When several numerals are used in one group, the space between numerals shall be not less than $\frac{1}{64}$ inch.

3.21.1.1 Markings.—The meter dial shall be legibly marked black on black with the following:

MS24332 (applicable part number)
Manufacturer's Part Number
Manufacturer's name or trademark
(Omit address)

3.21.1.2 Identification of product.—Equipment, assemblies, and parts shall be marked in accordance with MIL–STD–130.

3.22 Mounting screws.—The contractor shall furnish sufficient mounting screws for installation of the instrument in panels up to

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

3/16 inch thickness. They shall be No. 6–32 NC–2 round head brass machine screws having durable dull black finish.

3.22.1 Envelope.—An envelope, furnished by the contractor, containing mounting screws shall be packaged with each meter, and shall be marked with the following information:

"IMPORTANT"

"THIS ENVELOPE CONTAINS MOUNTING SCREWS"

3.23 Workmanship

3.23.1 General.—The voltmeter, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges.
b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.
c. Thoroughness of soldering, welding, brazing, painting, wiring, and riveting.
d. Alignment of parts and tightness of assembly screws and bolts; et cetera.

3.23.2 Cleaning.—The voltmeter shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.—The inspection and testing of the voltmeter shall be classified as follows:

a. Acceptance tests.—Acceptance tests are those tests accomplished on voltmeters submitted for acceptance under contract.
b. Qualification tests.—Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product. (See 4.6.)

4.2 Acceptance tests.—Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans A and B

4.2.1 Individual tests.—Each voltmeter shall be subjected to the following tests as described under 4.5 TEST METHODS:

a. Examination of product
b. Zero adjustment
c. Balancing
d. Accuracy
e. Damping

4.2.2 Sampling

4.2.2.1 Sampling Plan A.—Sampling plan A tests shall be performed on voltmeters selected at random in accordance with the following method:

Samples	*Lot*
1	1 to 50
2	51 to 200
3	201 to 1000
1	Each additional 1000

4.2.2.1.1 Sampling plan A shall consist of the following tests as described under 4.5 TEST METHODS:

a. Individual tests
b. Response time
c. Compass effect
d. Vibration
e. Temperature influence
f. Dielectric

4.2.2.2 Sampling Plan B.—Unless otherwise specified, three voltmeters shall be selected at random from the first 10 items produced under contract or order and subjected to the following tests as described under 4.5 TEST METHODS:

a. Individual tests
b. Sampling plan A
c. Heat effect at 71°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

d. Low temperature exposure
e. Sustained load
f. Humidity resistance
g. Shock
h. Mounting lugs

4.3 Rejection and Retest.—When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.1 Individual Tests May Continue.—For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.2 Defects In Items Already Accepted.—The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions.—Unless otherwise specified, the voltmeter shall be tested at room temperature of 25° ± 5°C.

4.4.1 Tapping.—Either the voltmeter or its support shall be tapped lightly in taking readings except when ascertaining the effect of friction after vibration.

4.5 Test Methods.

4.5.1 Examination of Product.—The voltmeter shall be inspected to determine compliance with this specification with regard to material, workmanship, and marking.

4.5.2 Zero Adjustment.—The voltmeter shall be checked for zero adjustment. The limits of the zero adjustment shall be as specified in 3.7.

4.5.3 Balancing.—The pointer shall be set to the center of the zero scale division with the meter in a vertical position. The instrument shall then be tilted 60° from the vertical in any direction and the position of the pointer shall not deflect more than the amount specified in 3.5.1.

4.5.4 Accuracy.—Each meter shall be calibrated to determine the accuracy of the instrument at room temperature. The accuracy may be checked at any reading on the scale and shall be within 2 percent of full scale deflection.

4.5.5 Damping.—Voltage sufficient to produce momentary full scale deflection shall be applied to the meter. The maximum momentary deflection shall be taken and recorded. After the pointer has come to rest, the steady deflection shall be taken and recorded. The damping factor shall be taken as the ratio of the steady deflection in angular degrees to difference between the maximum momentary deflection and the steady deflection in angular degrees.

4.5.6 Response time.—Voltage sufficient to produce momentary full scale deflection shall be applied to the meter. The length time, in seconds, required for the pointer to come to rest shall be noted. This test shall be repeated several times and the average value taken as the response time of the meter. The pointer is to be considered as having come to apparent rest when it has reached the actual rest point within ±1 percent of the scale length.

4.5.7 Compass effect.—The meter shall be rotated in a vertical plane about a short bar magnet type compass and with the nearest part of the indicator 5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the meter shall be held in position 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions the meter shall be in its normal upright mounting position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted.

4.5.8 Vibration.—Meters shall be subjected to vibration of 24 hours consisting of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

8-hour periods in each of the following 3 positions:

Position 1—Normal vertical mounting position.

Position 2—Rotated 90° counterclockwise in plane of mounting.

Position 3—Horizontal plane with face upward.

The construction of the vibrating platform shall be such as to give simple harmonic motion at a frequency of 400 cycles per minute and with an amplitude of 1/16 inch at the horizontal center line of the meter.

4.5.8.1 Meters shall also be mounted on a vibration stand giving a frequency between 500 and 2500 vibrations per minute of such a character that a point on the meter case will describe, in a plane inclined 45° to the plane of the face, a circle with a diameter of not less than 0.018 inch nor more than 0.020 inch. The meter shall be vibrated for a total of 2 hours at any desired frequency between 500 and 2500 vibrations per minute, but shall not be vibrated at any one frequency for more than 30 minutes. The meter shall be successively deflected to approximately 5 equidistant points on the scale during this test. In no case, however, shall the pointer be held at any given scale mark for more than 30 minutes.

4.5.8.2 At the conclusion of the vibration test the meter shall be connected to a course of potential and readings taken throughout the calibrated scale; the readings to be taken on both an ascending and descending scale, the latter to observe the effect of friction. The errors due to vibration shall not exceed the values specified in 3.5.11.

4.5.8.3 The condition of the pivots and bearings after the vibration tests shall be observed through a microscope of at least 40 diameters.

4.5.9 Temperature influence.—Meters shall be tested at 20°C above and below 25°C. Readings shall be taken at not less than 5 points (approximately equidistant) on the scale, including maximum scale value and the temperature influence computed. The temperature influence shall be considered one-half the percentage change in indication caused by the 20° departures from 25°C. If the influences above and below the reference temperature are not equal, the greater value shall be considered the temperature influence ascertained.

4.5.10 Dielectric.—The voltage as specified in 3.5.5 shall be applied between each terminal post and the metal panel upon which the instrument is mounted.

4.5.11 Heat effect at 71°C.—Voltmeters shall be placed in heated air and maintained at a temperature of 71°C for 16 hours. At the conclusion of this period and while still heated to this temperature, instruments shall be read throughout the calibrated scale. It shall be observed whether or not the meter indicates freely at all parts of the scale. Meters shall be allowed to cool at 25°C for 8 hours and another set of readings taken throughout the calibrated scale. The test shall be made 3 times successively to establish the effect of heat at 70°C.

4.5.12 Low temperature exposure.—The meter shall be subjected for a minimum period of 48 hours to a temperature of −65°C. The temperature shall then be raised to a temperature of −54°C and tested as specified for the accuracy test at low temperature to determine that no damage has resulted to the instrument.

4.5.13 Sustained load.—Voltmeters shall withstand an applied load corresponding to 100 percent of rated capacity for a period of 8 hours. The temporary zero shift of each meter shall be noted immediately upon withdrawal of load from that meter at the conclusion of the 8-hour period. Permanent errors in the indication and the permanent zero shift shall be ascertained 16 hours after removal of load.

4.5.14 Humidity resistance

4.5.14.1 Humidity conditioning.—One voltmeter with the cover in place and mounted in a vertical position shall be sub-

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

jected to the following humidity cycling test. At intervals during the test, the voltmeter should be examined for collection of moisture in the interior of the case, corrosion of metal parts, loosening or deterioration of the scale in any way, etc. The second voltmeter shall have its cover removed and shall be subjected to the humidity cycling tests, and inspection for the same features shall be conducted with the exception of collection of moisture within the case. The resistance of the internal multipliers, where used, shall be measured in each of the tests specified above.

4.5.14.2 Humidity cycling.—The voltmeter shall be placed in a chamber and kept for 6 hours at a temperature of 65°C, and relative humidity of 95 percent. After the required time, the voltmeter shall be cooled to normal ambient temperature for 18 hours in this atmosphere, taking precaution to prevent dripping on the meters and then tested for response, accuracy, damping factor and balance. The complete procedure shall be repeated a second time. The voltmeter shall operate satisfactorily at the conclusion of this test.

4.5.15 Shock.—The shock testing mechanism complying with JAN–S–44 is suitable for conducting this test. Voltmeters shall be mounted in their normal vertical mounting position and subjected to 10 shocks with an acceleration value of 50 each in a direction normal to the axis of the moving system at a rate of build-up and decay as defined by the shock-test device specified. The voltmeters shall then be rotated 90° counterclockwise in the plane of mounting and subjected to 10 shocks of the same magnitude and direction as previously. The voltmeters shall then be mounted in a horizontal plane, face upward, and subjected to 10 shocks of the same magnitude as at first but in a downward direction parallel to the axis of the moving system. Permanent error in indication shall be ascertained. Voltmeter mechanisms shall be examined for loosening of or damage parts. At the conclusion of this shock test, the meters shall be subjected to the vibration tests specified in 4.5.8.

4.5.16 Mounting lugs.—The instrument case shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting nut and attached to the pull strap in the stationary head of the machine. The specified load shall be applied to each lug for a period of 1 minute.

4.6 Qualification Testing

4.6.1 Qualification test samples.—The qualification test samples shall consist of eight voltmeters of each MS part number. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.6.2 Qualification required.—Prior to actual procurement, the product which this specification covers shall, pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.6.3. Qualification tests.—The qualification tests shall consist of all the tests specified in 4.5 TEST METHODS and

Test group 1 (all samples)
Examination of product
Zero adjustment
Balancing
Initial accuracy
Damping factor
Power consumption
Frequency influence
Response time
Compass effect
Temperature influence
Heat effect at 70°C.
Low temperature exposure
Insulation resistance
Insulation test

Test group 2 (2 samples)
Sustained load
Mounting lugs

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

Test group 3 (2 samples)
Humidity conditioning
Humidity cycling

Test group 4 (2 samples)
Shock
Vibration

Test group 5 (2 samples)
Supplementary tests, if necessary

5. PREPARATION FOR DELIVERY

5.1 General.—The requirements specified herein apply only to direct purchases by, or direct shipments to, the Government.

5.2 Packaging and packing.—The voltmeter shall be preserved, packaged, and packed for shipment in accordance with MIL–P–6064, group XI in a type I, class II container.

5.3 Marking of shipments.—Interior packages and exterior shipping containers shall be marked in accordance with MIL–STD–129. The nomenclature shall be as follows:

Voltmeter, 0–150 Volt, 330 To 1200 Cycle, Alternating Current

5.3.1 Reinspection marking.—The reinspection date markings shall be in accordance with ANA Bulletin 405.

6. NOTES

6.1 Use.—The voltmeters covered by this specification are intended for use in aircraft electrical systems.

6.2 Ordering data.—Procurement documents should specify the level of packaging and packing desired.

6.3 Provisions for qualification tests.—In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Bureau of Aeronautics, Navy Department, Washington 25, D. C.

Patent notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permision to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Transportation Corps
Navy—Bureau of Aeronautics
Air Force

Other Interest:
Army—Sig
Navy—MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

CHIEF, BUREAU OF NAVAL WEAPONS
ENGINEERING DIVISION
ATTN: CODE RREN-5
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

PLATE NO. 15419 (BACK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:25Z
Check the source to verify that this is the current version before use.